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# Algoritmi un programmēšanas metodes(English)(1),25/26-P QUIZ - T10 - Dijkstra's Algorithm

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**Marks** 5.00/5.00

**Grade** **10.00** out of 10.00 (**100%**)

## Question 1

Complete

Mark 1.00 out of 1.00

Dijkstra's Algorithm is used to solve the following problem:

Select one:

- ☐ a. All pairs shortest path
- ☒ b. Single source shortest path
- ☐ c. Knapsack problem
- ☐ d. None of the mentioned
- ☐ e. Sorting



**Question 2**

Complete

Mark 1.00 out of 1.00

What data structure would be preferable out of the following when implementing Dijkstra on an unweighted( $\text{weight}==1$  for all edges) graph?

Select one:

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- ☐ a. Stack
- ☐ b. Min Heap
- ☐ c. Binary Tree
- ☒ d. Queue

**Question 3**

Complete

Mark 1.00 out of 1.00

What is relaxation in Dijkstra algorithm?

Select one:

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- ☒ a. Testing whether a certain edge improves shortest path from source to given vertex.
- ☐ b. Finding weights for all edges for some vertex
- ☐ c. Testing whether we have visited a certain vertex already
- ☐ d. Performing a pass operation after successful completion of one round

**Question 4**

Complete

Mark 1.00 out of 1.00

Dijkstra's Algorithm can NOT be applied on which?

Select one:

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- ☐ a. Unweighted graphs
- ☒ b. Graphs with negative weight edges
- ☐ c. Directed and weighted graphs
- ☐ d. Undirected and unweighted graphs



## Question 5

Complete

Mark 1.00 out of 1.00

What is the preferable data structure to be used for Dijkstra algorithm for weighted graphs?

Select one:

- ☐ a. Array
- ☐ b. Linked List
- ☒ c. Heap
- ☐ d. Stack

◀ Dijkstra's Algorithm - barrera

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